Tutorial VI – Take-Off and Landing

Equally important to high-speed performance, is the aircraft's performance in the low-speed regime, i.e. during take-off and landing. Flight in these segments is often supported by deploying high-lift devices. This exercise shows the effect of leading and trailing edge flaps and the method for calculating the required take-off and landing distance.

Basics

- What is the effect on airfoil level when high-lift devices are deployed?
- Why does the take-off configuration of the high-lift devices differ from the landing configuration?
- Name the assumptions which are made when calculating runway lengths.

Calculation tasks

<u>Part 1: Take-off</u>

1) Determine the achievable lift coefficients due to trailing edge flaps ($\Delta C_{L,TE}$) and due to leading edge slats ($\Delta C_{L,LE}$). Use these results to calculate the maximum lift coefficient in take-off configuration $C_{L,max,TO}$ (flap lever is set to "2").
2) Also calculate the drag increments $\Delta C_{D,TE}$ and $\Delta C_{D,LE}$ in this case using the Roskam method. Use these results to calculate the total drag coefficient increment due to high-lift devices in take-off configuration $\Delta C_{D,HLD,TO}$.
3) Now calculate the <u>take-off</u> distance of the Airbus A320 at maximum take-off mass from an airport located at mean sea level. Break the calculation down to:
    a. Acceleration distance s_{acc} $(with \bar{T} \approx T_0 \cdot \frac{3}{4} \cdot \frac{5+BPR}{4+BPR})$
    b. Rotation distance s_{rot}
    c. Transition distance $s_{T,min}$

<u>Part 2: Landing</u>

4) Determine the achievable lift coefficients due to trailing edge flaps ($\Delta C_{L,TE}$) and due to leading edge slats ($\Delta C_{L,LE}$). Use these results to calculate the maximum lift coefficient in landing configuration $C_{L,max,L}$ (flap lever is set to "full").
5) Also calculate the drag increments $\Delta C_{D,TE}$ and $\Delta C_{D,LE}$ in this case using the Roskam method. Use these results to calculate the total drag coefficient increment due to high-lift devices in landing configuration $\Delta C_{D,HLD,L}$.
6) Finally, calculate the <u>landing</u> distance of the Airbus A320 at maximum landing mass from an airport located at mean sea level. Break the calculation down to:

a. Approach distance s_{L1}
b. Rotation distance $s_{L2,1}$
c. Deceleration distance s_{dec}

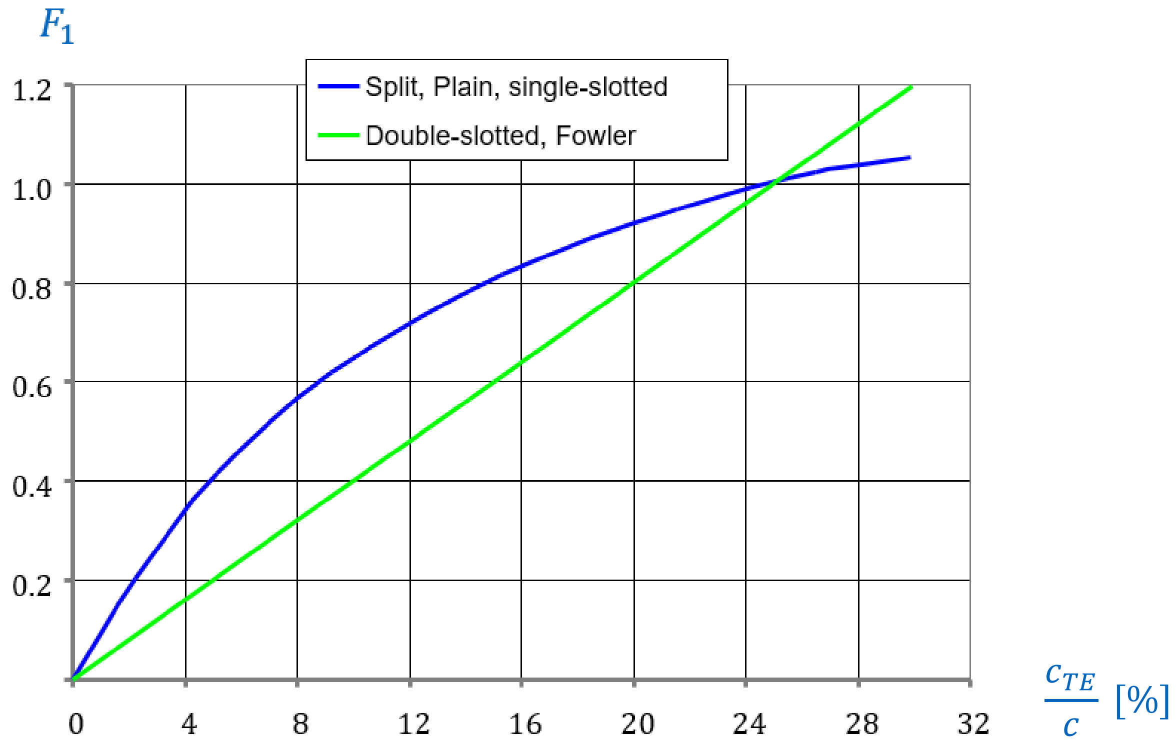
Assumptions and Additional Data

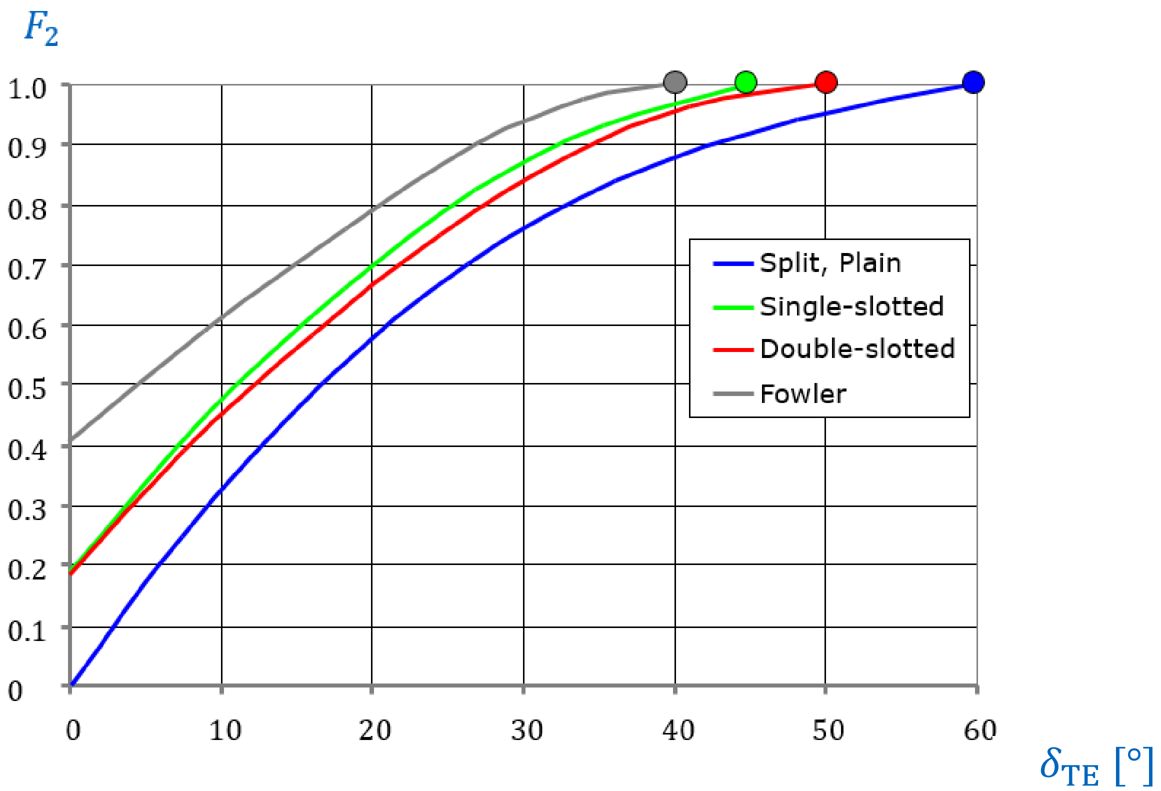
Maximum lift coefficient for clean configuration: $C_{L,max,clean} = 1.5$

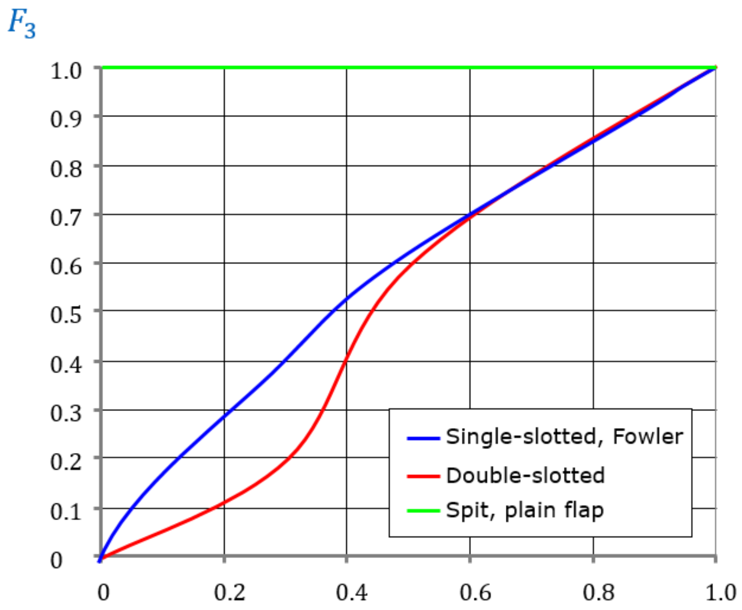
Flap system data:
Single-slotted trailing edge flap system, with chord: $c_{TE} = 1.05\ m$
Influences 74.7% of the wing reference area
Slat system data:
Slat leading edge system, with chord: $c_{LE} = 0.40\ m$ (assume that $c' = c + c_{LE}$)
Influences 72.8% of the wing reference area
Chord increase $\Delta c_{LE\ (TO)} = 0.05\ m$ & $\Delta c_{LE\ (L)} = 0.08\ m$
Take-off data:
Safety factor for take-off: $C = 1.15$ (according to MIL-Spec. or FAR)
Transition height: $h = 35\ ft$
Engine bypass ratio: $BPR = 5.5$
Rolling friction coefficient between tires and runway: $\mu_f = 0.02$
Zero-lift drag coefficient in clean configuration, low speed, sea level: $C_{D0|clean} = 0.018$
Drag coefficient increment due to landing gear: $\Delta C_{D,LG} = 1/4$ of zero-lift drag
Load factor at take-off transition: $n_z = 1.1$
Rotation duration: $\Delta t = 3\ s$
Wing parasite drag area at take-off: $f_W = 0.77$
Landing data:
Transition height: $h = 50\ ft$
Zero-lift drag coefficient in clean configuration, low speed, sea level: $C_{D0|clean} = 0.018$
Drag coefficient increment due to landing gear: $\Delta C_{D,LG} = 1/4$ of zero-lift drag
Rotation duration: $\Delta t = 3\ s$
Engine reverse thrust availability: $\eta_{TR} = 0.45$
Deceleration of landing gear braking system: $a = 3.9\ m/s^2$
Wing parasite drag area at landing: $f_W = 0.77$
General data:
Gravitational acceleration: $g = 9.81\ m/s^2$
Use the mean aerodynamic chord c_{MAC} as c in the high-lift calculations.
The relative airfoil thickness at MAC is not given; use this of the kink instead.

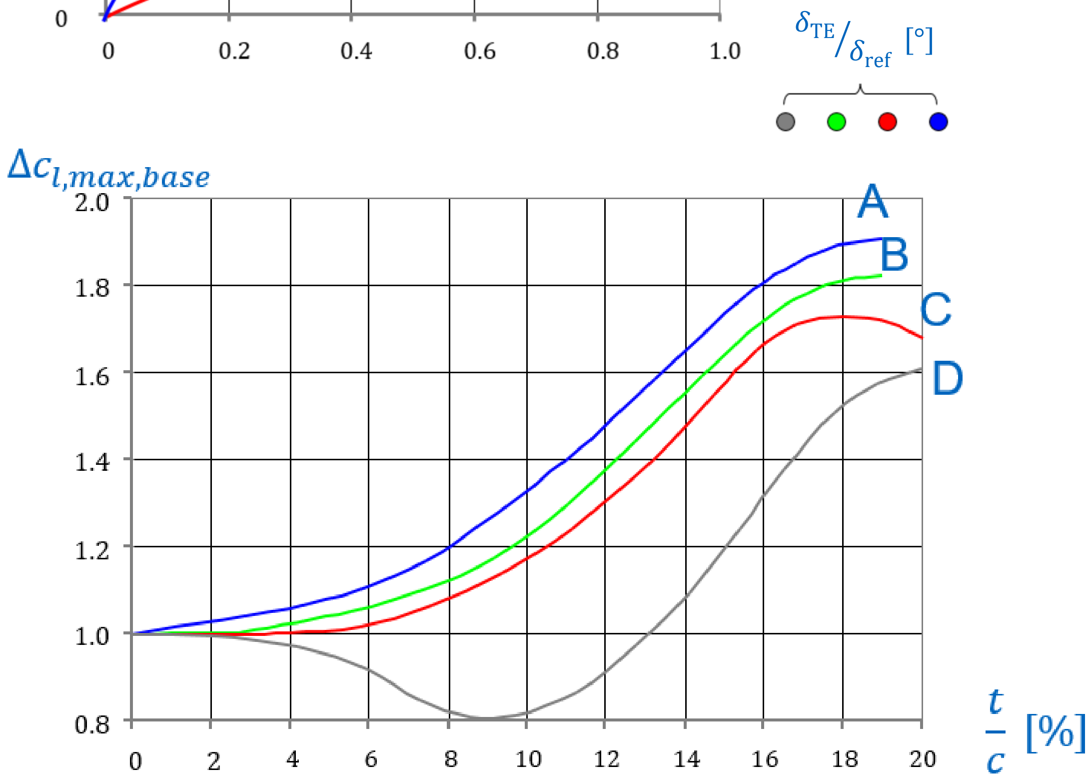
Flight phase	Lever position	Slat deflection	Flat deflection
Climb Cruise En-route Holding	0	0°	0°
Holding	1	18°	0°
Take-off	1 + F	18°	10°
Take-off Approach	2	22°	15°
Take-off Approach Landing	3	22°	20°
Landing (CFMI)	Full	27°	35°

Trailing edge system lift increment – Roskam method

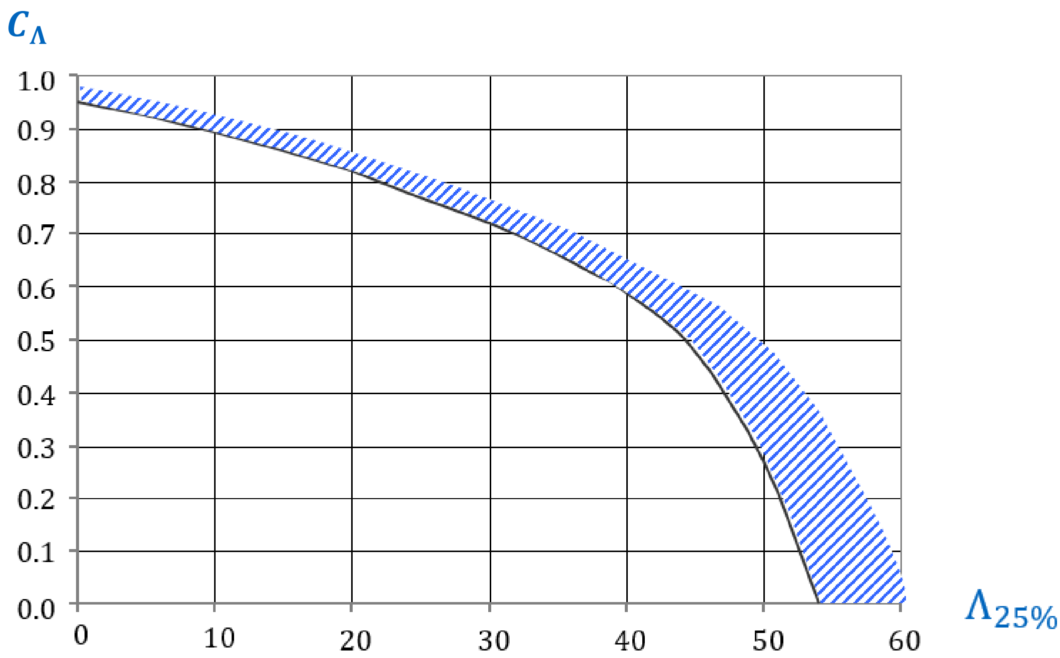




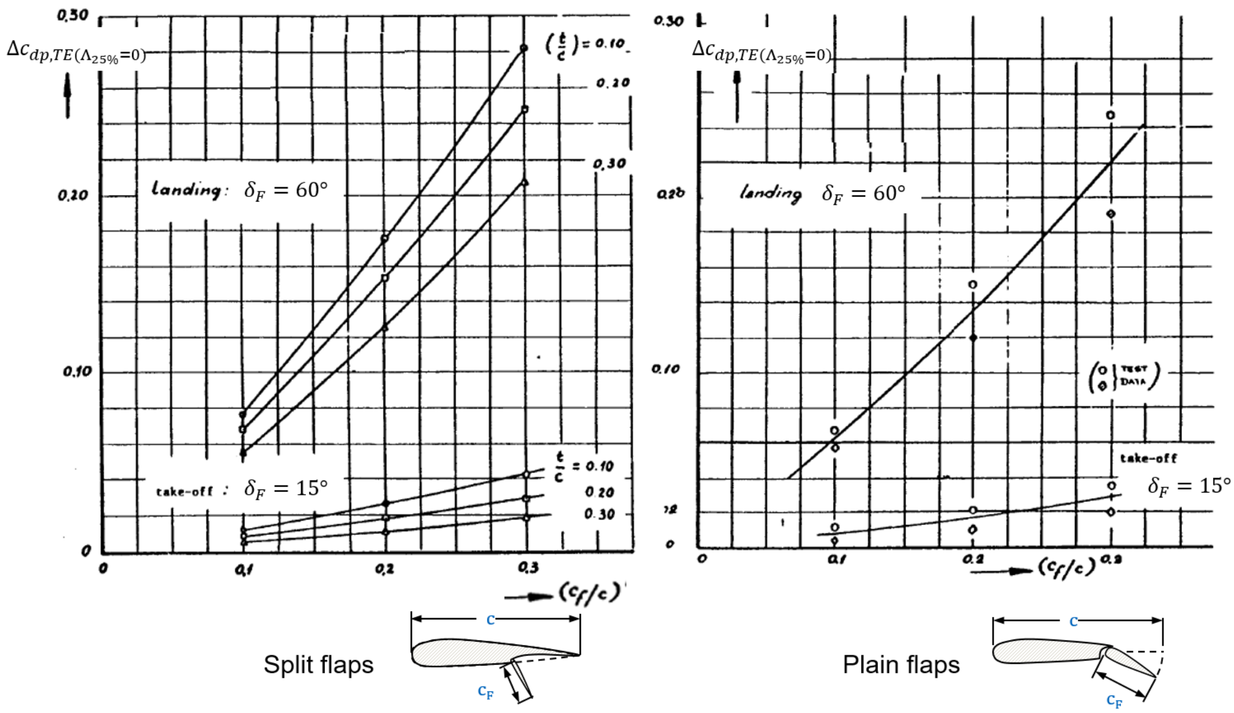


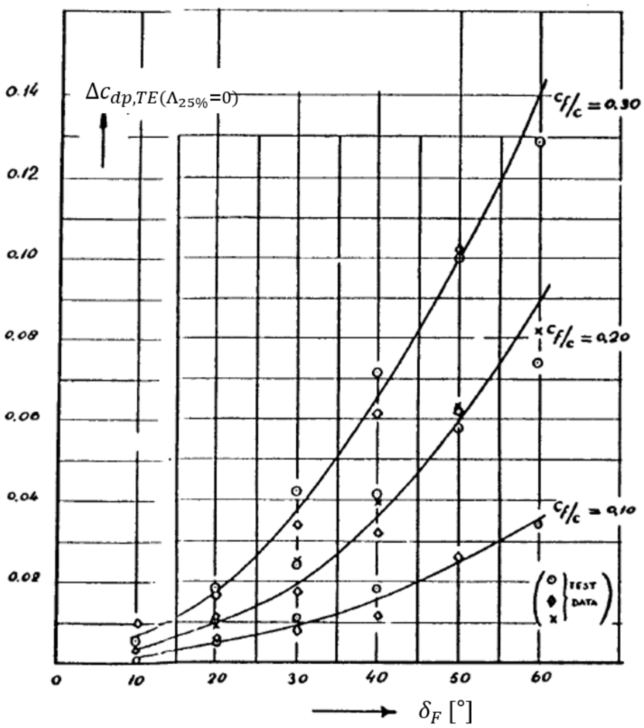


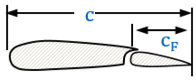
	Flap type	Airfoil type
A	Optimum double-slotted	NACA
B	Average double-slotted	NACA
	Fowler	any
C	Double slotted	NACA 6 series
	Single slotted	any
D	Split / plain	any



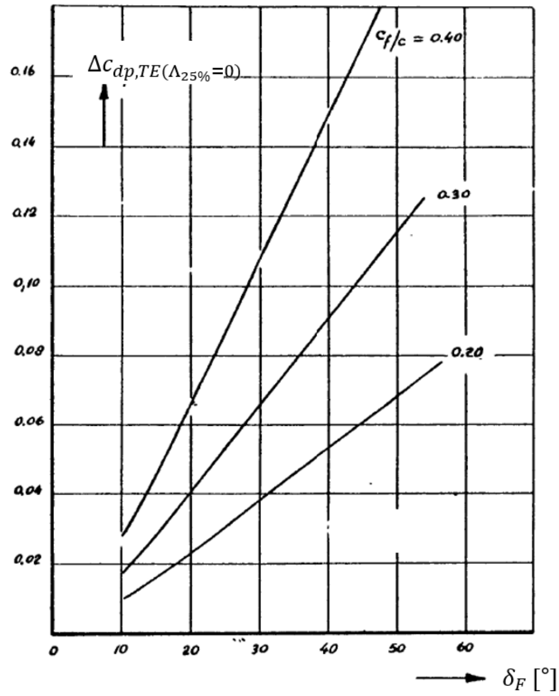
Trailing edge system drag increment – Roskam method

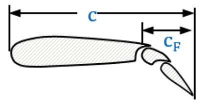




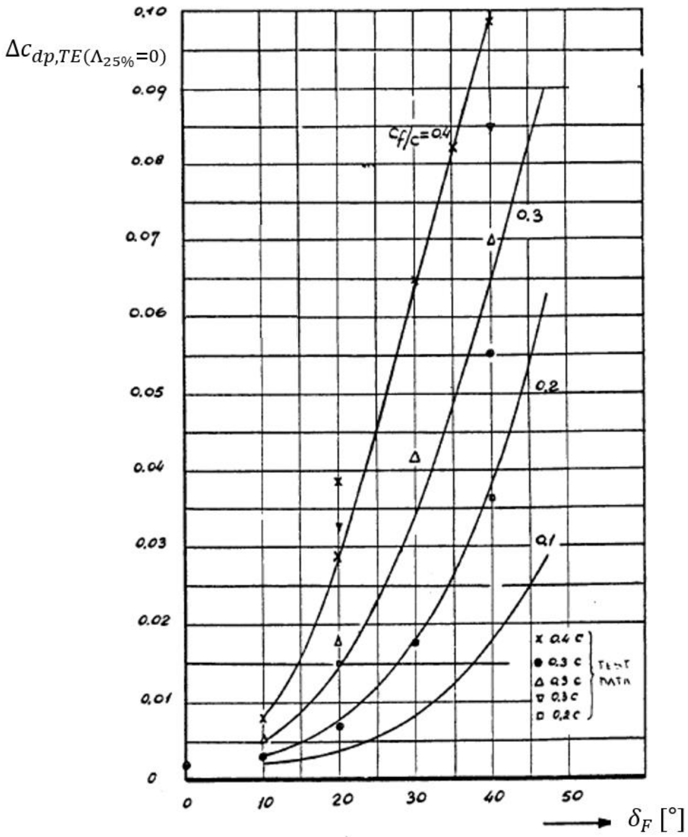


Single slotted flaps

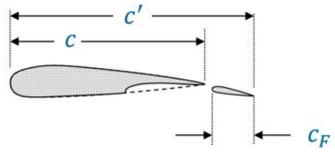




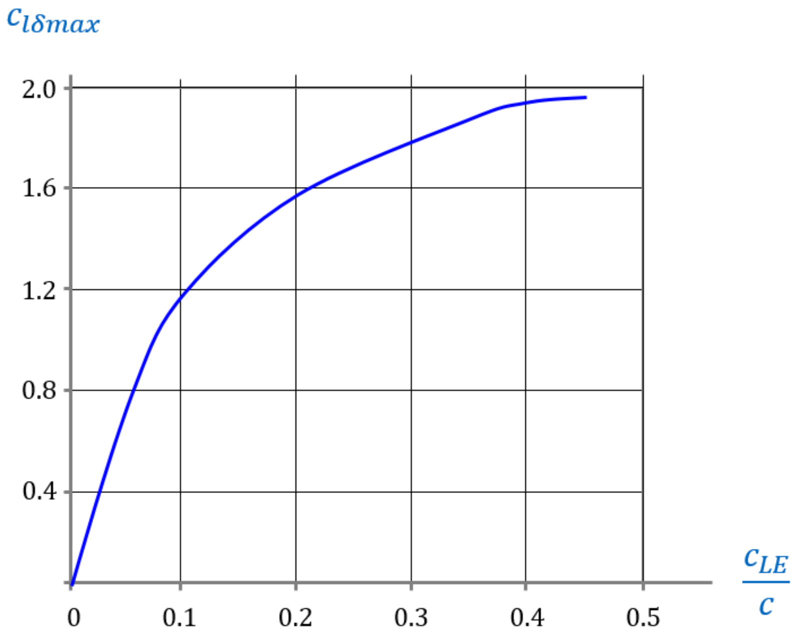
Double slotted flaps

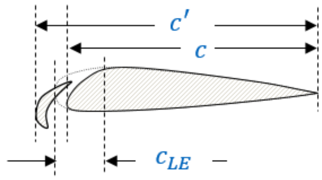


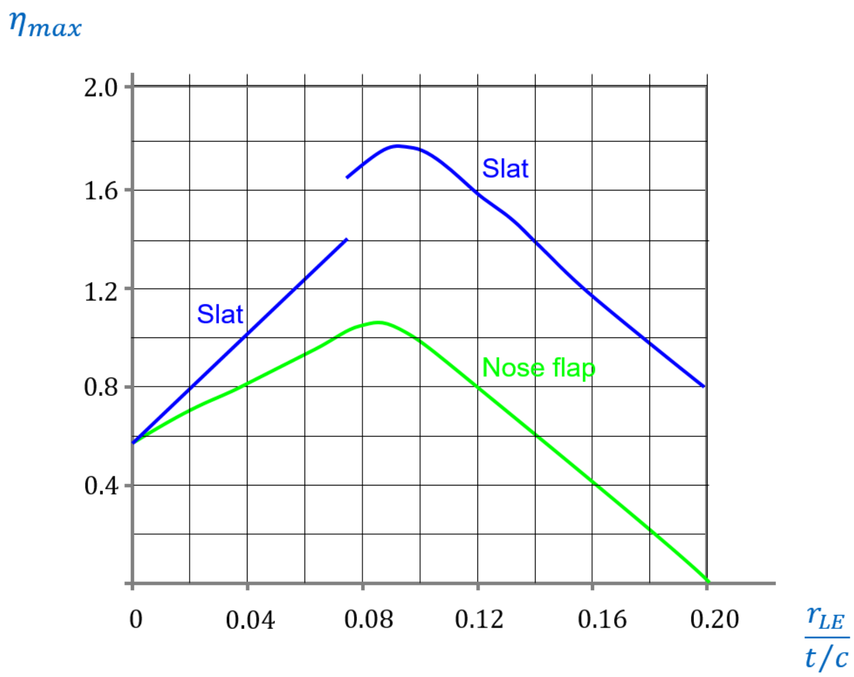
Fowler flaps



Leading edge system lift increment – Roskam method







Equations

$$\Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base}$$

$$\Delta C_{L,max(TE)} = \Delta c_{l,max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_\Lambda$$

$$\Delta c_{l,\max(LE)} = c_{l\delta max} \cdot \eta_{max} \cdot \delta_{LE} \cdot \left(\frac{c'}{c}\right) \cdot \pi/180$$

$$\Delta C_{L,max(LE)} = \Delta c_{l,\max(LE)} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos^2(\Lambda_{25\%})$$

$$\Delta C_{D,TE} = \Delta c_{dp,TE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

$$\Delta C_{D,LE} = \Delta c_{dp,LE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

For slats or Krüger flaps: $\Delta c_{dp,LE(\Lambda_{25\%}=0)} = C_{D0,W}\ (c + \Delta c_{LE})/c$

$$s_{\mathrm{acc}} = \frac{C^2 \cdot \frac{w_{TO}}{S_W}}{\rho \cdot g \cdot \left[C_{L,max,TO} \cdot \left(\frac{\bar{T}}{w_{TO}} - \mu_f\right) - \frac{C^2}{2} \cdot \left(\mathrm{C}_{D0,\mathrm{TO}} + \frac{{C_{L_{TO}}}^2}{3 \cdot \pi \cdot AR \cdot e_{TO}} - \mu_f \cdot C_{L,TO}\right)\right]}$$

$$s_{rot} = t_{rot} \cdot V_{TO}$$

$$s_{T,min} = \sqrt{\frac{2 \cdot h \cdot V_{TO}^2}{g \cdot (n_z - 1)}}$$

$$s_{L1} = \left(\frac{\bar{C}_L}{\bar{C}_D}\right) \cdot \left[h + \frac{1}{2 \cdot g} \cdot \left(V_{ap}^2 - V_{TD}^2\right)\right]$$

$$s_{L2,1} = \Delta t \cdot V_{TD}$$

$$s_{dec} = \frac{1.15^2}{\bar{a}} \cdot \frac{1}{C_{L,max,L}} \cdot \frac{1}{\rho} \cdot \frac{w_L}{S_W}$$

$$V_{ap} = 1.3 \cdot V_S$$

$$V_{TD} = 1.15 \cdot V_S$$

$$\bar{T} \approx T_0 \cdot \frac{3}{4} \cdot \frac{5 + BPR}{4 + BPR}$$

$$\bar{a} = a + \frac{T_0 \cdot \eta_{TR}}{w_L} \cdot g$$

$$\bar{C}_L = \frac{\bar{C}_{L,TD} + \bar{C}_{L,ap}}{2}$$